1. $T(n) = T(\frac{9n}{10}) + n$

Answer:

(a)

$$T(n) = T(\frac{9n}{10}) + n$$

$$T(\frac{9n}{10}) = T(\frac{9^2 n}{10^2}) + \frac{9n}{10}$$

$$T(\frac{9^2 n}{10^2}) = T(\frac{9^3 n}{10^3}) + \frac{9^2 n}{10^2}$$

...

$$T(\frac{9^{k-1} n}{10^{k-1}}) = T(\frac{9^k n}{10^k}) + \frac{9^{k-1} n}{10^{k-1}}$$

$$T(\frac{9^k n}{10^k}) = T(1)$$

$$\begin{array}{c} n \\ \downarrow \\ \frac{9n}{10} \\ \downarrow \\ \frac{9^2 n}{10^2} \\ \downarrow \\ \ldots \\ \downarrow \\ \frac{9^{k-1} n}{10^{k-1}} \\ \downarrow \\ 1 \end{array}$$

From the last equation, we have $\frac{9^k}{10^k} \cdot n = 1$

$n = (\frac{10}{9})^k$

$k = \frac{\log n}{\log(\frac{10}{9})}$

$k = \log_{(\frac{10}{9})} n$ Thus, the depth of the recursion is $\log_{(\frac{10}{9})} n$

The total cost of the recursion is the sum of the cost at each level for the depth of the recursion, which is $\log_{(\frac{10}{9})} n$.

Thus, the total cost is given by: $\sum_{k=0}^{\log_{(\frac{10}{9})} n} (\frac{9}{10})^k \cdot n$

$\sum_{k=0}^{\log_{(\frac{10}{9})} n} (\frac{9}{10})^k \cdot n = n \cdot \frac{1 - (\frac{9}{10})^{\log_{\frac{10}{9}} n + 1}}{1 - \frac{9}{10}}$

$n \cdot \frac{1 - (\frac{9}{10})^{\log_{\frac{10}{9}} n + 1}}{1 - \frac{9}{10}} = n \cdot \frac{1 - \frac{1}{n} \cdot \frac{9}{10}}{\frac{1}{10}}$

$T(n) = 10n - 9$

Thus, $T(n) = \Theta(n)\blacksquare$

(b)

$T(1) = 1$

$T(2) = T(\frac{18}{10}) + 2 = 3.8$

$T(3) = T(\frac{27}{10}) + 3 = 5.7$

$T(100) = T(\frac{900}{10}) + 100 = 190$

This begins to look like $2n$

Hypothesis: $T(k) \leq ck$ for all $k < n$

Substitute: $T(n) \leq c \cdot (\frac{9n}{10}) + n$

$T(n) \leq n(0.9c + 1)$

This is the desired form of $n \cdot c$

$0.9c + 1 \leq c$ for all $c \geq 10$

Thus, $T(n) \leq 10n$ for all $c \geq 10$

Therefore, $T(n) = \Theta(n) \blacksquare$

2. $S(n) = 9S(\frac{n}{3}) + n^2$

Answer:

(a)

Recursive call	**Tree Element**	**Number of Elements**	**Sum of Tree Elements**
$S(n) = 9S(\frac{n}{3}) + n^2$	n^2	1	n^2
$S(\frac{n}{3}) = 9S(\frac{n}{3^2}) + (\frac{n}{3})^2$	$(\frac{n^2}{9})$	9	n^2
$S(\frac{n}{3^2}) = 9S(\frac{n}{3^3}) + (\frac{n}{3^2})^2$	$(\frac{n^2}{81})$	81	n^2
$S(\frac{n}{3^{k-1}}) = 9S(\frac{n}{3^k}) + (\frac{n}{3^{k-1}})^2$	$(\frac{n^2}{3^{k-1}})$	3^{k-1}	n^2

$S(1) = \frac{n^2}{3^{2k}} = 1$
$3^{2k} = n^2$
$k \log 3 = \log n$
$k = \log_3 n$
The depth, k, is equal to $\log_3 n$ which means n^2 is summed $\log_3 n$ times, i.e. $n^2 \cdot \log_3 n$
Therefore, $S(n) = \Theta(n^2 \log n)$■

(b)

Inductive Hypothesis: $S(k) \leq ck^2 \log_3 k$ for all $k < n$
Substitute: $S(n) \leq 9c(\frac{n}{3})^2 \log_3(\frac{n}{3}) + n^2$
$S(n) \leq cn^2 \log_3(\frac{n}{3}) + n^2$
$S(n) \leq cn^2 \log_3 n - cn^2 \log_3 3 + n^2$
$S(n) \leq cn^2 \log_3 n - (cn^2 - n^2)$
$(cn^2 - n^2) \geq 0$ for all $c \geq 1$
Thus, $S(n) \leq cn^2 \log_3 n$ for all $c \geq 1$
Therefore, $S(n) = \Theta(n^2 \log n)$■